

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Condensed Matter Physics **L-T-P:** 3-0-0 **Credits:** 3

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Overview Lecture: What is condensed matter physics? Frontier areas in condensed matter physics. (1)

Crystal Structure of Lattice: Lattice, Symmetry, Unit cell, Bravais Lattices, Reciprocal Lattice, Brillouin zone, Crystal Planes and Directions, X-ray Diffraction (Laue condition, Bragg's law) (6)

Electronic properties of crystalline solids: Free-electron theory, Weak periodic potential, Nearly free electron theory, Bloch's Theorem, Tight-binding approximation, Band dispersion, Density of states, Distinction of Metals, Semiconductors and Insulators from electronic band structure (10)

Lattice Vibration and its Effects: Phonons due to lattice vibrations: 1D mon- and di-atomic chains; Phonon's role in heat capacity (Einstein and Debye theory); Electron-phonon interaction; Effective attractive interaction, Formation of Cooper pairs, Coherent state of Cooper pairs: the BCS ground state of superconductivity; Experimental Detection of Phonons: Raman scattering and Neutron scattering. (10)

Transport Properties: Boltzmann transport theory, Scattering due to random potential and phonons, Length scales for different transport regimes: Diffusive and Ballistic, Weak localization correction to conductivity. (7)

Magnetism: Diamagnetism and Paramagnetism, Some examples of cooperative magnetism. (3)

Introductory Discussion on Frontier Topics (any two): Mesoscopic physics, Quantum Hall effect, Topological systems, Magnetic systems, Light-matter interaction, Unconventional superconductors, Strongly correlated electronic systems. (5)

Reference Books:

1. Solid State Physics by N. W. Ashcroft and N. D. Mermin.
2. The Oxford Solid State Basics by S. H. Simon.
3. Solid State Physics by C. Kittel
4. Solid State Physics by H. Ibach and H. Lüth
5. Condensed Matter Physics by M. P. Marder
6. Modern Condensed Matter Physics by S. M. Girvin and K. Yang
7. Electronic Transport in Mesoscopic Systems by Supriyo Datta

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Overview Lecture: What is condensed matter physics? Frontier areas in condensed matter physics.	1

2.	Crystal Structure of Lattice: Lattice, Symmetry, Unit cell, Bravais Lattices, Reciprocal Lattice, Brillouin zone, Crystal Planes and Directions, X-ray Diffraction (Laue condition, Bragg's law)	6
3.	Electronic properties of crystalline solids: Free-electron theory, Weak periodic potential, Nearly free electron theory, Bloch's Theorem, Tight-binding approximation, Band dispersion, Density of states, Distinction of Metals, Semiconductors and Insulators from electronic band structure	10
4.	Lattice Vibration and its Effects: Phonons due to lattice vibrations: 1D mon- and di-atomic chains; Phonon's role in heat capacity (Einstein and Debye theory); Electron-phonon interaction; Effective attractive interaction, Formation of Cooper pairs, Coherent state of Cooper pairs: the BCS ground state of superconductivity; Experimental Detection of Phonons: Raman scattering and Neutron scattering.	10
5.	Transport Properties: Boltzmann transport theory, Scattering due to random potential and phonons, Length scales for different transport regimes: Diffusive and Ballistic, Weak localization correction to conductivity.	7
6.	Magnetism: Diamagnetism and Paramagnetism, Some examples of cooperative magnetism.	3
7.	Introductory Discussion on Frontier Topics (any two): Mesoscopic physics, Quantum Hall effect, Topological systems, Magnetic systems, Light-matter interaction, Unconventional superconductors, Strongly correlated electronic systems.	5
Total number of hours		43